

Oluwasemilore Adelaja

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PROFESSIONAL SUMMARY

Data scientist and machine learning engineer specialising in deep learning; Mechanical Engineering undergraduate at the University of Lagos. I build what most people import: a raw-NumPy deep learning framework with every gradient hand-derived, a bitboard chess engine, and a transformer from the paper up.

EDUCATION

University of Lagos

B.Eng. Mechanical Engineering — First Class Honours (4.69/5.0)

Lagos, NG

Oct. 2021 – Present

Relevant Coursework: Engineering Computing, Numerical Methods, Engineering Statistics

EXPERIENCE

Engineering Intern

Chevron Nigeria Limited

Mar. 2026 – Sept. 2026

Lagos, Nigeria

AI Engineer

Aegis AI — AfriHackBox 2025 Continental Hackathon (Winner)

2025

Remote

- Served as the AI engineer for a phishing-detection app that won the AfriHackBox 2025 continental hackathon
- Engineered a hybrid system pairing an offline “Sentry” model with an online “Oracle” model to enable real-time offline analysis

PROJECTS

MR & N: Deep Learning Framework in Pure NumPy | *Python, NumPy* | [GitHub](#)

- Built a deep learning framework in base Python and NumPy with every gradient derived by hand and no autograd: dense and convolutional layers, dropout, momentum-based gradient descent, learning-rate scheduling, and a range of activation and cost functions
- Rewrote the array-based engine as a doubly-linked-list, class-based core, making the module graph more modular and extensible

Chess Programming: Bitboard Engine and Unity Front End | *C#, Unity* | [GitHub](#)

- Wrote a chess engine from scratch with bitboard state, generating fully legal moves directly through pin and check analysis rather than make/unmake filtering
- Shipped BondFish 1.0–1.4, adding one search technique per version (minimax, alpha-beta pruning, quiescence search); verified move generation with a perf harness against known node counts

LectureKit: Lecture Recording to Typeset PDF | *Python, Whisper, LaTeX* | [GitHub](#)

- Built a five-stage offline pipeline turning a lecture recording into a formatted PDF with a timecoded transcript, abstractive summary, and glossary of the technical terms actually spoken
- Chained faster-whisper transcription, distilbart map-reduce summarisation, fuzzy glossary matching, and Jinja to LaTeX rendering; runs on a laptop CPU with no API keys, and every stage caches so a failed run resumes

Transformer: “Attention Is All You Need” from Primitives | *Python, PyTorch* | [GitHub](#)

- Implemented a decoder-only transformer bottom-up from raw PyTorch tensors, a single self-attention head through to the full stack, trained on tiny_shakespeare with a character-level tokenizer

StockBot: Next-Day Direction Prediction on SPY | *Python, scikit-learn* | [GitHub](#)

- Framed prediction as binary next-day direction rather than a price target, training a random forest on daily SPY data with features from rolling windows at five horizons (2 to 250 days)
- Computed trend features on shifted data to eliminate lookahead leakage and scored on precision; served the model through an API behind a web front end

WRITING

Neural Networks Demystified ([Medium](#), [code](#)): rebuilding neural networks in NumPy, from perceptrons to a NumPy-only MNIST classifier

The Art of Secrecy ([Medium](#), [code](#)): classical and modern cryptography in Python, including a full modular Enigma simulation

EXTRACURRICULARS

Two-time winner and one-time competition MVP — ULES UNILAG Inter-departmental Engineering Quiz Competition

TECHNICAL SKILLS

Languages: Python, C#, SQL, LaTeX

Machine Learning: Deep Learning, Reinforcement Learning, Computer Vision, NLP, Transfer Learning

Libraries & Data: NumPy, PyTorch, scikit-learn, Pandas, SciPy

Engineering & Tools: Backpropagation from scratch, Bitboards, Alpha-beta search, Monte Carlo Tree Search, Self-attention, Unity, Git